

Final Usability Report
Managing Information Overload: Examining TripAdvisor's Effectiveness in Supporting
Complex Trip Planning

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Introduction

Traveling is a fun and fulfilling experience where individuals venture to familiar and new places. Despite the joy its outcome brings, the process of coordinating travel is quite complex as there are countless components that need to align together seamlessly, which can be frustrating to plan. Trip planning requires lots of time, information processing and decision making, which can be challenging to manage all at once. Therefore, travelers who use online travel platforms expect that the site's organization and guidance will alleviate this taxing process. Considering this, my usability test examines whether users can manage the complexity of planning a full trip on TripAdvisor without experiencing information overload.

With an emphasis on complexity and cognitive load, I am addressing the problem of users losing track of their trip details and ultimately, getting confused in the process. From the perspective of user experience, if users feel overwhelmed or frustrated while using TripAdvisor, it will negatively impact their planning efficiency. They may struggle to navigate and process the information, which reduces their choice selection and overall confidence in cultivating a smooth voyage. Ultimately, if users feel displeased with their experience on TripAdvisor, they may abandon their process and use a competitor's site instead. This problem directly affects the business goals that are centered around user retention, user conversion and overall revenue. TripAdvisor wants to retain its users, hoping that they stay engaged with the platform or reuse it for multiple separate trips. If the platform fails to be intuitive, this results in a reduction of user retention. Similarly, TripAdvisor wants its users to complete actions such as creating an account or writing a review; however if users are dropping before doing so, conversion rates drop as well. These both influence general revenue as users may leave TripAdvisor to continue their planning on competitor's sites, driving engagement away from them.

My research goals are to measure task complexity, information processing efficiency, decision clarity and user satisfaction/confidence. In alignment with user goals, these areas define the necessary components of a planning experience that facilitates users seamlessly through a stressful process. Travelers desire to complete tasks quickly, to process information fully, to make decisions without hesitation and to be confident/satisfied with it all. In alignment with business needs, TripAdvisor wants to make sure its position in the market is strengthened,

keeping them competitive in a growing industry. TripAdvisor's selling point is being a one-stop-shop, so this usability test can provide clarity on its design.

This selling point speaks to my decision in choosing to examine TripAdvisor which is an exhaustive travel guidance platform. This widely used interface features extensive user-generated content such as reviews and rankings, a plethora of booking options that ranges from hotels to restaurants, personalized traveler insights and social media integrations. As it is designed to support all facets of a trip, it is both content-rich and complex which is extremely relevant to my research goals. Additionally, it is promoted to support the issues of management and decision-making which are also important to my interest.

Methodology

I administered a remote, moderated task-based test. Although I followed a script, I also asked open-ended probing questions throughout the sessions. After the main session, participants were then asked to complete a post-test survey. To give greater context for the study design (as encouraged from peer feedback), participants were each given the same trip scenario: A 3-day leisure trip to Montego Bay, Jamaica from August 1st-3rd, 2025 with their significant other. There were four tasks in total (in this respective order): 1) book a convenient round-trip flight, 2) book a modern hotel that has at least 4 stars, 3) book a day activity on Day 2, and 4) book a nearby restaurant after the activity. Each task was followed up with questions that assessed their decision-making criteria and experience. After all tasks were completed, participants were asked recall questions that assessed their cognitive load and retention. Finally, in the post-test survey, participants answered questions that assessed their satisfaction, decision confidence, cognitive load and perceived ease of use.

I selected this combination of methods as it provided a great balance of reality and simulation, allowing for objective observation and subjective user reflection. When travelers are booking trips, they usually do so individually on their own devices. Through remote testing, I was able to replicate this as participants completed the test in their natural environment on their respective devices. A moderated format grants me the opportunity to observe participants' body language, hear their real-time thoughts and probe immediately to gather clarifications, which is necessary

for the complex process they were completing. A task set not only gives participants a set goal in mind to keep them aligned but also it guarantees that the session will include a full, end-to-end trip where participants will need to compare, navigate and decide. Although using the same structured scenario does not capture the full range of travel use cases, it gave me a baseline to be able to completely compare results from different participants. Finally, a post-test survey helped me to receive user reflections and ratings that indicate their satisfaction and confidence levels.

TripAdvisor's target audience includes a wide range of users who come from all backgrounds and age groups, with a slight focus on millennials and generation Z. Its platform is accessible to all income and educational backgrounds. With budgets that can be either tight or luxurious, users' travel purposes consist of leisure, business, or both. Majority of their users are tech-savvy and are quite familiar with navigating online interfaces. They are independent and sociable as they prefer to book their own trips, yet are highly influenced by peer recommendations or other reliable information. My target participants for this study closely resembled TripAdvisor's target audience. As the general user goal of TripAdvisor is to book something for a trip, I wanted to select participants that matched their current market as it also aligned with my main research objective.

For recruiting participants, I utilized my own personal network of family, friends and peers. I created a participant screener that was disseminated amongst multiple forums and chats, receiving more than 18 responses. This screener included questions that assessed demographic information, platform preferences, travel habits, comfortability with online booking sites and familiarity with TripAdvisor. Despite personal recruitment, I aimed for a good degree of diversity in user habits amongst my 5 users to ensure generalizability. My criteria for selecting participants from the screener was also again based on reflecting TripAdvisor's target audience. For demographic considerations, I prioritized those within the age ranges of 18-24 and 25-34; however, beyond this I had no specifications on cultural backgrounds or genders - aiming for a mixture. For behavioral considerations, I chose participants based on both their user journey and product use frequency. I considered how often they used TripAdvisor, the types of trips they have planned before and also if they were a first-time or occasional user of TripAdvisor. My selected participants all had wavering degrees of technical proficiency, travel proficiency and

TripAdvisor proficiency. As they came from my personal network, all participants were English-speakers, resided in a similar geographic location and possessed cultural habits that resemble that of modern Westerners. There were no applicable special considerations or accessibility needs that I needed to consider.

Data Collection and Analysis

As my methodology was mixed in design, I similarly collected both qualitative and quantitative data. The qualitative data I collected were observational notes and user feedback. This type of data enables me to perceive the participants' decision-making, information processing and reflections which is crucial to assess complexity and cognitive load. Observations helped me to gauge signs of mental distress whether through prolonged sighs or faces of confusion. User feedback gave me verbal confirmation of the frustrations or thoughts that I had observed. The quantitative data I collected were task completion time, cognitive load (backtracking rate + information retention score) and confidence/satisfaction rate. Task completion time points to a user's ability to process information and/or navigate the platform. Therefore, a longer task completion time can indicate a low task efficiency or even a low system efficiency. Cognitive load is calculated through an averaging of their backtracking rate and information retention score. As advised from previous feedback, I will provide more details about this metric. I defined a backtrack as any time a user clicks the back arrow, starts their selection or process over, reopens a previous choice or navigates to a different tab ("Hotels" -> "Things to Do"). To uncover the backtracking rate, it will be the number of backtracks over the number of total pages the user visited. For their information retention score, if the user can correctly answer the recall question such as the hotel's name, they get a full 1 point. A 0.5 is noted if they get part of the answer correct and a 0 for the incorrect or lack of answer. This is averaged across the total number of full points for recall questions. I combined this backtracking rate with the cognitive score and averaged it over the 4 tasks. A high cognitive load indicates that the user has trouble retaining information and may point to overwhelm. Confidence/satisfaction rates reveal users' feelings and point to ease of use. If these rates are low, it could indicate mental drain due to information overload or a lack of support.

To draw meaningful insights, I looked at the participant's quantitative values and then compared that with my qualitative notes for context and validation. For example, when users had high confidence/satisfaction scores and also verbalized the site's ease of use, it confirmed that complexity was not an issue. When conducting my analysis, I appreciated this mixture of information as it enabled me to paint a greater picture. The combination of objective and subjective data assisted me in discovering pain points and identifying usability issues.

The most common theme that emerged from my testing is that TripAdvisor is an easy and intuitive platform that provides users with lots of information in an organized and digestible format. The word map below (thanks to Loop11) compiled by participants' text entry showcases the general sentiment.

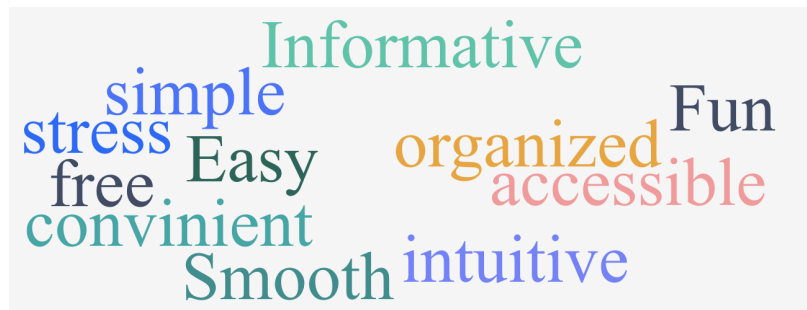


Fig. 1 - Loop11 Generated Word Map

This notion is backed both by the post-test survey results (displayed in Figure 2) and the closing remarks of each participant.

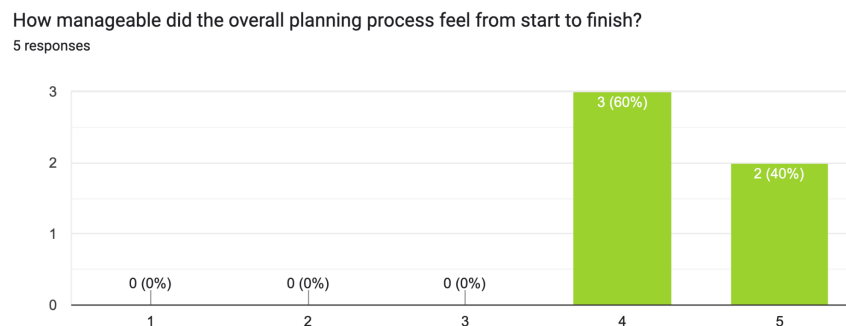


Fig. 2 - TripAdvisor's Manageability

80% of the participants' overall satisfaction and confidence ratings ranked above 85% which also communicates users' feelings towards TripAdvisor. Trends that I have observed were low average cognitive scores despite varying degrees of backtracking, high confidence in final choices, and the consistent pain point of trying to remember location details for proximity to

restaurants or activities. I would say that besides 3 out of 5 users showing signs of frustration for the lack of location assistance, other issues that arose were unique to the user. Finally, it is worth noting that although the tasks of choosing a hotel and excursion on average had the highest task completion time (5:53 mins and 5:43 mins, respectively), these choices were ranked as the most confident as displayed in Figure 3. This suggests that as users spend more time browsing and selecting their choices, they will be more confident in those decisions.

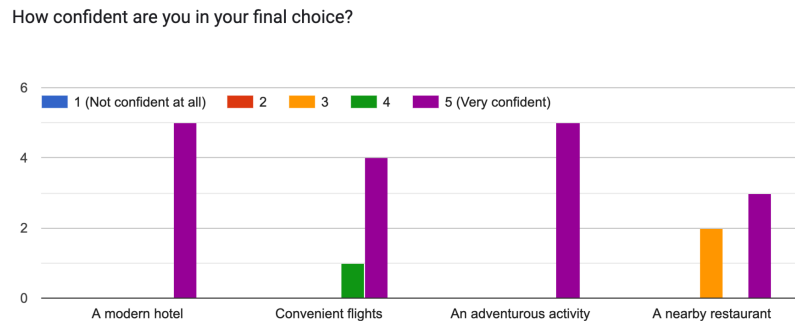


Fig. 3 - Confidence in Final Choices

Findings and Recommendations

The key usability issues that were identified during testing were a lack of planning support, low perceived credibility and unexpected behavior. These issues speak to cognitive load, navigation intuitiveness and general user trust. As I mentioned before, cognitive load directly impacts planning efficiency and low navigation intuitiveness adds on to the friction of system use. If a user feels that they cannot fully trust the information provided by TripAdvisor, they may seek the advice from other platforms and not engage with specific features on the platform. This decrease of user trust leads to lower engagement and leaves a negative impression on TripAdvisor's brand perception.

60% of users commented on TripAdvisor's lack of location features which would remind users of the proximity of their choices. For instance, if a user has chosen their hotel and is now looking at excursions, they currently cannot identify how close the activity is to their residence without using the aid of other websites like Google Maps (which one user had to do) or trying to estimate it themselves. Participant 3 commented that through their use of Google Maps, they felt that

TripAdvisor was “not a complete tool.... It [TripAdvisor] got me 80% of the way”. Because this issue was prominent and triggered greater frustration, I would denote this to be a severity of 3. It disrupted the user flow by slowing them down and forcing them to take more mental notes. My actionable recommendation would be to incorporate a mini-tool or small notification that calculates the distance between the user’s hotel location and the activity or restaurant location and reminds the user of it. Another solution could be to place recommendations for a user at the top that is based off of their hotel location. For instance, the list of options could be under a header that exclaims “Close to your Hotel”. This is practical as TripAdvisor contains lots of sections that are influenced by popularity or other rankings. Again, because this will prevent users from leaving TripAdvisor, I believe this is an important change to increase user conversion and further facilitate the process.

Participant 1 remarked how in their process for finding a nearby restaurant, they felt that the TripAdvisor photos were “outdated and old”. Due to this perception, they felt a platform such as TikTok would be better suited for restaurant recommendations as it could provide tangible videos that are more recent. Participant 1 would then rank their restaurant choice as 60% confident and satisfied in the post-test survey. As this directly impacted both their confidence and satisfaction, I would denote this to be a severity of 2. Photos are quite important when assessing restaurant food, so this limits the decision clarity for users and slows them down. An actionable recommendation would be to incorporate short video clips as a part of restaurant reviews, so that users can feel more informed. This is a practical change as TripAdvisor contains extensive user-generated content as it expands to incorporate more social media-like features. This is important to improve the product as increased user engagement and trust also increases positive brand perception.

Finally, Participant 5 experienced difficulty with the search bar as the entry restarted their initial progress. Instead of showing listings of a specific activity, it redirected the participant to the main destination page. Participant 5 immediately expressed verbal confusion and frustration as they had to restart their excursion search. Given this experience, they noted a $\frac{2}{3}$ difficulty for activity selection despite a $\frac{1}{3}$ difficulty for all other tasks. As this slowed down the user and did not meet typical search engine expectations, I would denote this a severity of 3. An actionable

recommendation is to fix the engine so that it showcases results that pertain to the context of the user's tab. For example, if a user is under the "Things to Do" tab, the search results showcase links to specific excursions only. This recommendation would require the leg work of the engineers more so than the designers, so I will say that the practicality is slightly diminished. Despite lower practicality, this change will reduce user frustration and thus impact user satisfaction.

Overall, I would say that these findings align with the needs of stakeholders as it is important to support the user from start to finish of their journey. Any gaps that are identified need to be addressed as the perceived lack of support from TripAdvisor could lead to users abandoning their process and seeking to continue it elsewhere. With the focus on retention, conversion and revenue, business leaders and product owners alike would want to enact these improvements immediately to solidify their position in the travel industry. To accommodate varying levels of technical expertise, I would further utilize the strengths of charts and graphs in order to convey the key findings and demonstrate the correlations between my UX findings and typical business terms.

Ethical Considerations and Limitations

To ensure this study remained ethical, there were many precautions taken throughout. For participant recruitment, I maintained frequent and in-depth communication about the study at all points of the process. During screener distribution, I highlighted the screener's purpose of filtering participants for an upcoming remote usability session that would take approximately 45 minutes and notified respondents that upon completion of the survey they could potentially be selected. Upon selection, all chosen participants received an email that expressed their selection for the study and information including the study's time duration, software choice and testing format (tasks followed by a survey). Additionally, the email asked for permission to audio/video record the session, ensuring that these recordings would be shared anywhere and would only be seen by me. If permission was granted, participants would download the software's extension to be used for the study, which further confirmed permission. In the beginning of the actual test, participants were made aware that they can decline to answer questions and choose to take

breaks. Finally, all collected data was securely stored on the software's database and any personal data such as names were removed/anonymized to maintain confidentiality.

Ethical challenges to consider included power dynamics and emotional discomfort. Because I recruited participants from my personal network, they could have felt obligated to participate. In recognizing this, I emphasized how the study was purely voluntary and there were no social consequences for declining to participate. There were instances in which survey respondents declined to move further, so this communicates that this challenge was not a huge one. Being watched in a seemingly serious context could intimate participants and cause discomfort. To mitigate this, I stressed to participants that I was not testing them but rather testing TripAdvisor. Additionally, I reminded them that there were no right or wrong answers, so they should be their natural selves.

Several biases from different parts of my research process could have influenced my findings. From my participant selection, there is a degree of selection bias that is present as I was limited to those in my personal network. As addressed in my methodology, despite the social network proximity, I aimed to incorporate a wide range of diversity that still emulated TripAdvisor's target audience. Although my 5 users may differ from each other in multiple ways, this group cannot be fully representative of the general population. The rest of the biases derive from my test design. Because it follows a moderated format, my research is vulnerable to the Hawthorne effect where users may act differently because they know they are being watched. The steps I took to reduce this effect included operating a remote session in the user's natural environment, using a software where the user could not see my screen when completing tasks, and prompting users to behave normally and treat this trip as if they were actually going. During the session, participants may succumb to social desirability bias, or saying what they think I want to hear. Participants may think that they should have a good experience with TripAdvisor compared to a negative one. To reduce this bias, I emphasized that I was testing the product, not them and I also made sure to balance questions to include both ends of the spectrum ("surprising or confusing"). Finally, observer bias will sometimes come into play quite unconsciously. To mitigate this, I enacted use of quantitative metrics which are predefined and objective as well as recorded sessions in order to revisit and confirm findings.

Limitations in my research design are due to sample size, time constraints and task design. I only completed this study with 5 participants who come from a somewhat similar demographic as they belong to my personal networks. For future tests, I would recruit from a wider user base that further increases diversity across age groups and more. As gathered from my findings, all of my participants spread their planning out over time. Asking participants to plan 4 different aspects of a trip all in one session does not simulate their real use and efforts. For future tests, I would implement a longitudinal study that breaks up tasks over time to give them more flexibility and feel more natural. Finally, my task design of producing a fake trip may have not carried as much weight for users than an actual upcoming trip would. The emotional and cognitive stress is diminished drastically as participants know in the back of their minds that this session is still make-believe. For future tests, I could follow users who are planning an actual trip and utilize a diary study to capture their thoughts. Logistical challenges to note purely lied in methodological changes. I conducted a pilot study with my original script and tasks and had to not only change the ordering of the tasks but also some wording. For instance, I originally had the hotel as the first task but I recognized that logically, travelers book their flights first to even know where to find their hotel. Additionally, I changed the verbiage of a question to ask for “two adjectives” rather than just “two words”. These challenges were minor, yet impactful.

Executive Summary

According to the findings of my usability study, users sometimes struggle to manage the complexity of planning a full trip due to factors such as the lack of platform support for location and low perceived credibility; however, this does not have an immense negative influence on their general experience or final choices. Users, on average, feel adequately guided through their trip-planning process, describing TripAdvisor to be “easy” and “stress-free”. Because of TripAdvisor’s vast source of information, users are expected to be facilitated in all aspects, which expresses their frustration in not finding helpful information about location proximity. Additionally, users did not feel that they could fully trust and believe reviews as some data appeared to be outdated. Actionable insights that can increase users’ trust, users’ willingness to use TripAdvisor for all of their needs (user conversion) and users’ likelihood of returning to TripAdvisor for future trips (user retention) include adding integrated location services and

updating user-generated content such as videos and photos. Next steps for TripAdvisor would be to work on making these changes and then conducting greater usability testing afterwards.

All in all, TripAdvisor does a pretty good job of maintaining its competitive advantage amongst its competitors like Booking.com; however, it can continue to refine its efforts to increase the amount of returning users and general revenue due to increased usage.

Citations -

Easy user testing tool for data-driven UX. Loop11. (2025, February 10). <https://www.loop11.com/>